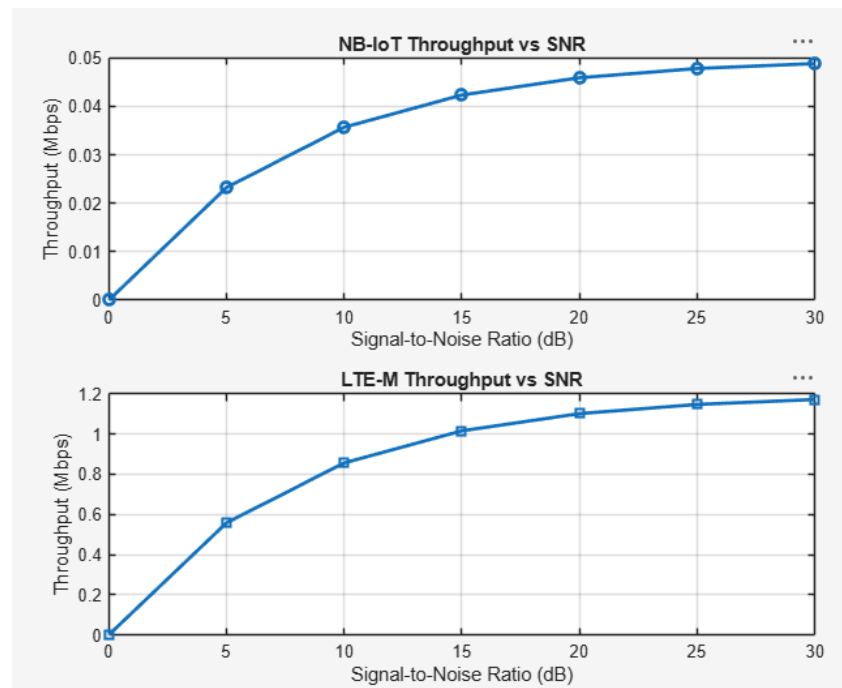


Ex.No.12: NB-IoT vs LTE-M Throughput Simulation

Objective 1: Matlab Code and Output

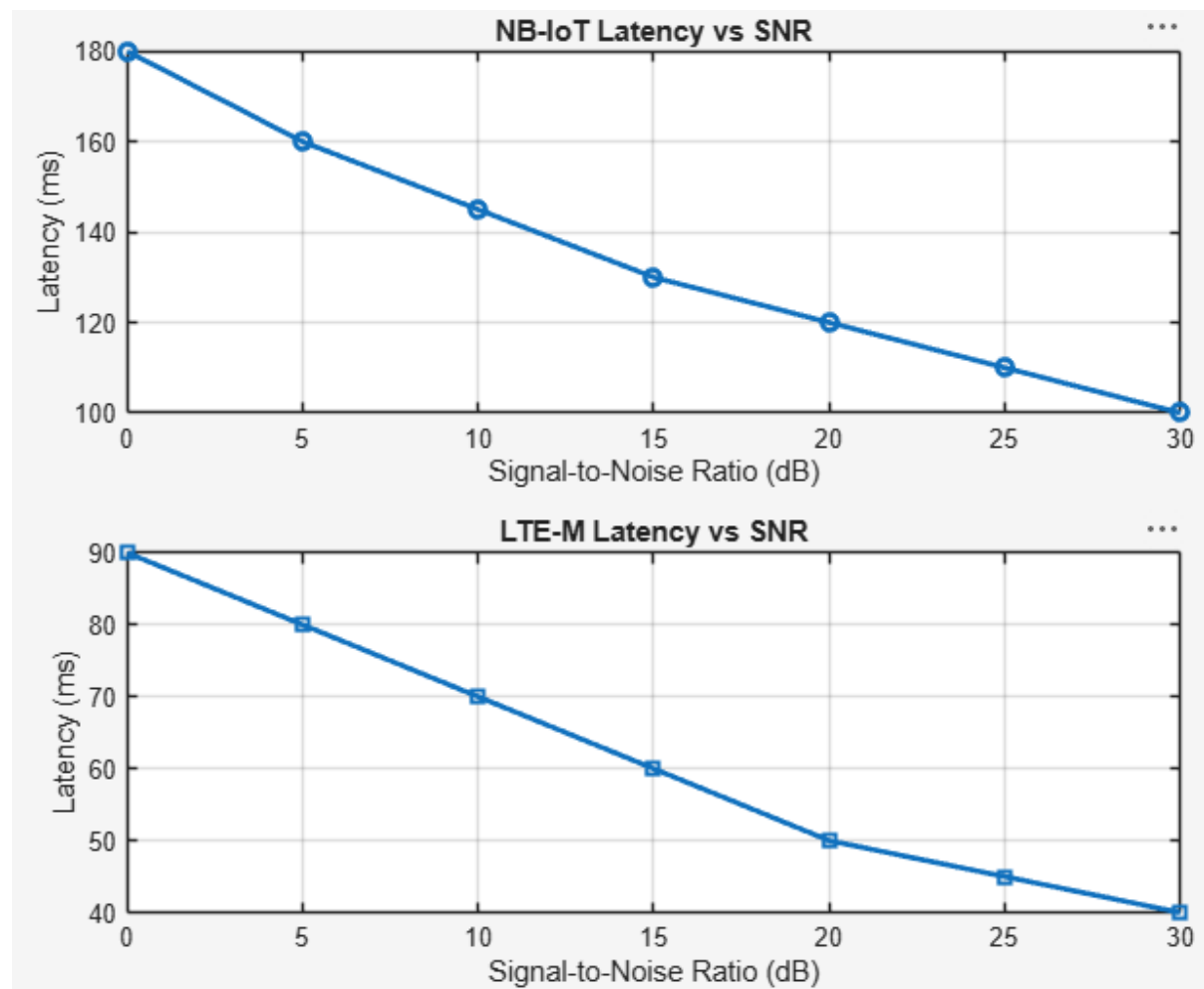
/MATLAB Drive/slotb12.m

```
1   clc;
2   clear;
3   close all;
4   % SNR values (dB)
5   SNR = 0:5:30;
6   % Simulated Throughput (Mbps)
7   NB_IoT = 0.05*(1-exp(-SNR/8));
8   LTE_M = 1.2*(1-exp(-SNR/8));
9   disp('SNR (dB)    NB-IoT(Mbps)    LTE-M(Mbps)')
10  disp([SNR' NB_IoT' LTE_M'])
11  figure
12  % ----- Graph 1 -----
13  subplot(2,1,1)
14  plot(SNR,NB_IoT,'-o','LineWidth',2)
15  title('NB-IoT Throughput vs SNR')
16  xlabel('Signal-to-Noise Ratio (dB)')
17  ylabel('Throughput (Mbps)')
18  grid on
19  % ----- Graph 2 -----
20  subplot(2,1,2)
21  plot(SNR,LTE_M,'-s','LineWidth',2)
22  title('LTE-M Throughput vs SNR')
23  xlabel('Signal-to-Noise Ratio (dB)')
24  ylabel('Throughput (Mbps)')
25  grid on
```



Objective 2: Matlab Code and Output

```
27 % SNR values (dB)
28 SNR = 0:5:30;
29 % Simulated Latency (ms)
30 NB_IoT = [180 160 145 130 120 110 100];
31 LTE_M = [90 80 70 60 50 45 40];
32 disp('SNR(dB)    NB-IoT Latency(ms)    LTE-M Latency(ms)')
33 disp([SNR' NB_IoT' LTE_M'])
34 figure
35 % ----- Graph 1 -----
36 subplot(2,1,1)
37 plot(SNR,NB_IoT,'-o','LineWidth',2)
38 title('NB-IoT Latency vs SNR')
39 xlabel('Signal-to-Noise Ratio (dB)')
40 ylabel('Latency (ms)')
41 grid on
42 % ----- Graph 2 -----
43 subplot(2,1,2)
44 plot(SNR,LTE_M,'-s','LineWidth',2)
45 title('LTE-M Latency vs SNR')
46 xlabel('Signal-to-Noise Ratio (dB)')
47 ylabel('Latency (ms)')
48 grid on
```



Objective 3: Matlab Code and Output

```
50 % SNR values (dB)
51 SNR = 0:5:30;
52 % Simulated Packet Delivery Ratio (%)
53 NB_IoT = [82 86 89 92 95 97 99];
54 LTE_M = [88 91 94 96 98 99 100];
55 disp('SNR(dB)    NB-IoT PDR(%)    LTE-M PDR(%)')
56 disp([SNR' NB_IoT' LTE_M'])
57 figure
58 % ----- Graph 1 -----
59 subplot(2,1,1)
60 plot(SNR,NB_IoT,'-o','LineWidth',2)
61 title('NB-IoT Packet Delivery Ratio vs SNR')
62 xlabel('Signal-to-Noise Ratio (dB)')
63 ylabel('Packet Delivery Ratio (%)')
64 grid on
65 % ----- Graph 2 -----
66 subplot(2,1,2)
67 plot(SNR,LTE_M,'-s','LineWidth',2)
68 title('LTE-M Packet Delivery Ratio vs SNR')
69 xlabel('Signal-to-Noise Ratio (dB)')
70 ylabel('Packet Delivery Ratio (%)')
71 grid on
```

